

```
In [2]: text='    abc def ghi    '
text.strip()
```

```
Out[2]: 'abc def ghi'
```

```
In [3]: #Using double quotes in the string is not allowed.
mystr = "My favourite TV Series is "Game of Thrones""
```

```
Cell In[3], line 2
    mystr = "My favourite TV Series is "Game of Thrones""
                                   ^
SyntaxError: invalid syntax
```

```
In [4]: list1=[]
print(type(list1))
```

```
<class 'list'>
```

```
In [12]: list2 = [10,30,60] # List of integers numbers
```

```
In [13]: list3 = [10.77,30.66,60.89] # List of float numbers
```

```
In [14]: list4 = ['one','two' , "three"] # List of strings
```

```
In [15]: list4 = ['one','two' , "three"] # List of strings
```

```
In [16]: list6 = [100, 'Asif', 17.765] # List of mixed data types
```

```
In [11]: list7 = ['Asif', 25 ,[50, 100],[150, 90] , {'John' , 'David'}]
```

```
In [10]: len(list6) #Length of List
```

```
Out[10]: 3
```

```
In [17]: mylist = ['one' , 'two' , 'three' , 'four' , 'five' , 'six' , 'seven' , 'eight']
```

```
In [18]: mylist[0:3]
```

```
Out[18]: ['one', 'two', 'three']
```

```
In [19]: mylist[:3]
```

```
Out[19]: ['one', 'two', 'three']
```

```
In [20]: mylist[:2]
```

```
Out[20]: ['one', 'two']
```

```
In [21]: mylist[-3:]
```

```
Out[21]: ['six', 'seven', 'eight']
```

```
In [22]: mylist[-2:]
```

Out[22]: ['seven', 'eight']

```
In [23]: #add remove and changes items  
mylist
```

Out[23]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']

```
In [24]: mylist.append('nine')  
mylist
```

Out[24]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']

```
In [26]: mylist.insert(9, 'ten')  
mylist
```

Out[26]: ['one',
 'two',
 'three',
 'four',
 'five',
 'six',
 'seven',
 'eight',
 'nine',
 'ten',
 'ten']

```
In [27]: mylist.insert(1, 'ONE')  
mylist
```

Out[27]: ['one',
 'ONE',
 'two',
 'three',
 'four',
 'five',
 'six',
 'seven',
 'eight',
 'nine',
 'ten',
 'ten']

```
In [28]: mylist.remove('ONE')  
mylist
```

Out[28]: ['one',
 'two',
 'three',
 'four',
 'five',
 'six',
 'seven',
 'eight',
 'nine',
 'ten',
 'ten']

```
In [29]: mylist.pop()  
mylist
```

The history saving thread hit an unexpected error (OperationalError('database or disk is full')).History will not be written to the database.

```
Out[29]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten']
```

```
In [30]: mylist.pop(8)  
mylist
```

```
Out[30]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'ten']
```

```
In [31]: del mylist[7]  
mylist
```

```
Out[31]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'ten']
```

```
In [32]: #change value of the string  
mylist[0] = 1  
mylist[1] = 2  
mylist[2] = 3  
mylist
```

```
Out[32]: [1, 2, 3, 'four', 'five', 'six', 'seven', 'ten']
```

```
In [33]: mylist
```

```
Out[33]: [1, 2, 3, 'four', 'five', 'six', 'seven', 'ten']
```

```
In [34]: mylist.clear() # Empty List / Delete all items in the List  
mylist
```

```
Out[34]: []
```

```
In [35]: #copy List  
mylist
```

```
Out[35]: []
```

```
In [38]: mylist = ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']  
mylist
```

```
Out[38]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [40]: mylist1 = mylist # Create a new reference "mylist1"  
mylist1
```

```
Out[40]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [41]: mylist
```

```
Out[41]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [42]: id(mylist) , id(mylist1)
```

```
Out[42]: (2546390869888, 2546390869888)
```

```
In [43]: id(mylist) , id(mylist1)
```

```
Out[43]: (2546390869888, 2546390869888)
```

```
In [45]: mylist2 = mylist.copy()  
mylist2
```

```
Out[45]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [46]: id(mylist2)
```

```
Out[46]: 2546390805248
```

```
In [48]: mylist[0] = 1  
mylist
```

```
Out[48]: [1, 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [49]: mylist1
```

```
Out[49]: [1, 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [50]: mylist2
```

```
Out[50]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [ ]: # mylist1 will be also impacted as it is pointing to the same list
```

```
In [ ]: #Copy of list won't be impacted due to changes made on the original list
```

```
In [52]: list1 = ['one', 'two', 'three', 'four']  
list2 = ['five', 'six', 'seven', 'eight']
```

```
In [53]: list3 = list1 + list2 # Join two lists by '+' operator  
list3
```

```
Out[53]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [54]: list1.extend(list2) #Append list2 with list1  
list1
```

```
Out[54]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [55]: #List membership  
list1
```

```
Out[55]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [56]: 'one' in list1
```

```
Out[56]: True
```

```
In [57]: 'ten' in list1
```

```
Out[57]: False
```

```
In [58]: if 'three' in list1:
        print('present')
        else:
        print('not present')
```

present

```
In [59]: #reverse and soert List
        list1.reverse() # Reverse the List
        list1
```

Out[59]: ['eight', 'seven', 'six', 'five', 'four', 'three', 'two', 'one']

```
In [61]: list1=list1[::-1]
        list1
```

Out[61]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']

```
In [62]: mylist3 = [9,5,2,99,12,88,34]
        mylist3.sort() # Sort List in ascending order
        mylist3
```

Out[62]: [2, 5, 9, 12, 34, 88, 99]

```
In [63]: mylist3 = [9,5,2,99,12,88,34]
        mylist3.sort(reverse=True) # Sort List in descending order
        mylist3
```

Out[63]: [99, 88, 34, 12, 9, 5, 2]

```
In [64]: mylist4 = [88,65,33,21,11,98]
        sorted(mylist4) # Returns a new sorted List and doesn't change original
```

Out[64]: [11, 21, 33, 65, 88, 98]

```
In [65]: mylist4
```

Out[65]: [88, 65, 33, 21, 11, 98]

```
In [66]: #Loop through List
        list1
```

Out[66]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']

```
In [68]: for i in list1:
        print(i)
```

one
two
three
four
five
six
seven
eight

```
In [69]: for i in enumerate(list1):
        print(i)
```

```
(0, 'one')  
(1, 'two')  
(2, 'three')  
(3, 'four')  
(4, 'five')  
(5, 'six')  
(6, 'seven')  
(7, 'eight')
```

```
In [71]: #all and any  
l1 = [1,2,3,4,0]
```

```
In [72]: all(l1)
```

```
Out[72]: False
```

```
In [73]: any(l1)
```

```
Out[73]: True
```

```
In [74]: l2 = [1,2,3,4,True,False]
```

```
In [75]: all(l2)
```

```
Out[75]: False
```

```
In [76]: any(l2)
```

```
Out[76]: True
```

```
In [79]: l3 = [1,2,3,True]
```

```
In [80]: all(l3)
```

```
Out[80]: True
```

```
In [81]: any(l3)
```

```
Out[81]: True
```